Founding Engineer Take-Home

Lawyer Identity Resolution

The problem

We have name strings of advocates, AOR (Advocates on Record), and Senior Advocates extracted from Supreme Court documents. We need to resolve these into canonical identities.

Problem 1 — Name variation

These names have both valid and invalid variations.

Valid variations (should merge into one person):

- **Misspellings.** "Gopal Sankaranarayanan" can appear with 20+ different misspellings — "Gopal Sankanarayan", "Gopal Sankanarayanan", "Gopal Sankarayanan", etc.
- **Abbreviations.** "Rajan K. Chourasia" can be abbreviated as "R. K. Chaurasia", "R. Kumar Chaurasia", "Raja Kumar Chourasia", and so on.

Invalid variations (should NOT merge — these may be different people):

- The common surname "Aggarwal" is also spelled "Agarwal" and "Agrawal" — but these can be different people, not necessarily the same person spelled differently.
- "Goyal" and "Goel" are both valid spellings — but they may be different people.

The task is to find the genuine misspellings and abbreviations and merge them into a single canonical name (typically choosing the most frequent spelling), while not over-merging.

Problem 2 — Designation resolution

Every lawyer has one of three designations: **Advocate**, **AOR**, or **Senior Advocate**. Here too, there are valid and invalid variations.

Valid: An advocate "abc" might have been an AOR in 2023, but from 2024 she was designated Senior and now appears as "Senior Advocate". Her current status should be Senior.

Invalid: An advocate "xyz" was mistakenly typed as "Senior" in a few documents (registry error) and never actually got promoted. He should remain an Advocate.

You need to detect these patterns and assign each lawyer their correct current designation.

Problem 3 — Master lists

- We do **not** have a master list of all advocates who appear in the Supreme Court.
- We **do** have a master list of Advocates on Record (`aor_master.csv`).

- We have a master list of Senior Advocates (`sr_advocate_master.csv`), but it is **incomplete** — Senior Advocates can also be designated by High Courts, and once designated they remain Senior even when they appear at the Supreme Court. Absence from this list does not mean someone is not a Senior Advocate.

The data

`/data/appearances.csv` — each row is one lawyer appearance in one Supreme Court hearing.

Column	Notes
`appearance_id`	Unique row ID
`raw_name`	The lawyer's name as recorded in the document. Noisy.
`designation_raw`	Designation as recorded, with formatting variation
`case_id`	Groups appearances by case. Rows sharing a `case_id` are co-counsel in the same hearing.
`hearing_date`	ISO date
`appearing_for`	Petitioner, Respondent, etc.

`/data/aor_master.csv` — official AOR registry.

`/data/sr_advocate_master.csv` — official Senior Advocate registry (incomplete, see Problem 3).

What to deliver

1. **A design document** describing your approach. Cover how you decompose the problem, which methods you use where (deterministic rules, statistical methods, LLMs, etc.) and why, how you use the master lists, how you validate quality with no ground truth labels, and what you'd be most worried about getting wrong.
2. **A working prototype** that solves at least a meaningful slice of the problem. You do not need to cover everything end-to-end — pick something you can do well and be explicit about what you skipped.
3. **Output** as a CSV: `appearance_id`, `cluster_id`, `canonical_name`, plus any confidence scores or flags you compute.
4. **Code we can run**, with a short README.